Note

The energy of the orbiting electrons is measured in electron volts (eV).

10.1.2 Classification of materials

The classification of solids into insulators, metals, and semiconductors can be explained with the help of the energy band diagram.

i) Insulators

The energy band structure of insulators is shown in Figure 10.2(b). The valence band and the conduction band are separated by a large energy gap. The forbidden energy gap is approximately 6 eV in insulators. The gap is very large that electrons from valence band cannot move into conduction band even on the application of strong external electric field or the increase in temperature. Therefore, the electrical conduction is not possible as the free electrons are not available for conduction and hence these materials are called insulators. Its resistivity is in the range of 10^{11}–$10^{19}\,\Omega$m.

ii) Conductors

In condutors, the valence band and conduction band overlap as shown in Figure 10.2(c). Hence, electrons can move freely into the conduction band which results in a large number of free electrons available in the conduction band. Therefore, conduction becomes possible even at low temperatures. The application of electric field provides sufficient energy to the electrons to drift in a particular direction to constitute a current. For condutors, the resistivity value lies between $10^{-2}\,\Omega$m and $10^{-8}\,\Omega$m.

iii) Semiconductors

In semiconductors, there exists a narrow forbidden energy gap $\left(E_g < 3eV\right)$ between the valence band and the conduction band (Figure 10.2(d)). At a finite temperature, thermal agitations in the solid can break the covalent bond between the atoms (covalent bond is formed due to the sharing of electrons to attain stable electronic configuration). This releases some electrons from valence band to conduction band. Since free electrons are small in number, the conductivity of the semiconductors is not as high as that of the conductors. The resistivity value of semiconductors is from $10^{-5}\,\Omega$m to $10^{6}\,\Omega$m.

Note

In semiconductors, electrons in the valence band are bound electrons which cannot move. Hence, they cannot contribute for conduction.

When the temperature is increased further, more number of electrons are promoted to the conduction band and they increase the conduction. Thus, we can say that the electrical conduction increases with the increase in temperature. In other words, resistance decreases with increase in temperature. Hence, semiconductors are said to have negative temperature coefficient of resistance. The most important commonly used elemental semiconducting materials are silicon (Si) and germanium (Ge). The values of forbidden energy gap for Si and Ge at room temperature are 1.1 eV and 0.7 eV respectively.

10.2 TYPES OF SEMICONDUCTRORS

10.2.1 Intrinsic semiconductors

A semiconductor in its pure form without any impurity is called an intrinsic semiconductor. Here, impurity means

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